ISA Lab Manual

October 9, 2025

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Part I

Lab Exercise: ISA Profiling

1 Introduction to the Lab

In this exercise, you will run pre-compiled benchmarks to analyze the distribution of instructions they execute. You will use the `sim-profile` simulator for this purpose. The goal is to understand the characteristics of different programs and compare the PISA and Alpha ISAs. All commands should be run from the `/SimpleScalar/simplesim-3.0/` directory.

1.1 Part 1: Profiling ALPHA Benchmarks

In this part, you will compile the simulators for the **ALPHA ISA** and run a suite of larger benchmark programs.

1.1.1 Step 1: Configure the Simulator for ALPHA

1. **Clean the directory** to remove any previous builds.

```
make clean
```

2. **Configure for ALPHA** to set up the build environment for the ALPHA ISA.

```
make config-alpha
```

3. **Build the simulators** by compiling the source code.

```
make
```

1.1.2 Step 2: Run the Benchmark Suite

Execute the following commands. After each one, find the `sim_inst_class_prof` section in the output to find the statistics needed for Table 1.

```
./sim-profile -iclass ../benchmarks/anagram.alpha ../benchmarks/words < ../
    benchmarks/anagram.in
```

- `-iclass`: Tells the simulator to profile the instruction class distribution.
- `../benchmarks/words`: An argument to the anagram program (the dictionary file).
- `< ../benchmarks/anagram.in`: Redirects the file `anagram.in` to be the standard input.

```
./sim-profile -iclass ../benchmarks/compress95.alpha < ../benchmarks/compress95
    .in
```

```
./sim-profile -iclass ../benchmarks/go.alpha 50 9 ../benchmarks/2stone9.in
```

```
./sim-profile -iclass ../benchmarks/cc1.alpha -O ../benchmarks/1stmt.i
```

1.1.3 Step 3: Data Collection and Analysis

Record your results in the table below and answer the questions that follow.

Table 1: Instruction Profile for ALPHA Benchmarks

Benchmark	**Total Insts**	**Load %**	**Store %**	**Uncond. Branch %**	**Cond. Branch %**	**Integer Compute %**	**Floating Pt. Compute %**
anagram.alpha							
go.alpha							
compress.alpha							
gcc.alpha							

Analysis Questions For each benchmark, answer the following based on your data:

1. Is the benchmark memory intensive (high load/store %) or computation intensive?
2. Is the benchmark using mainly integer or floating-point computations?
3. What percentage of instructions are conditional branches? Based on this, how many instructions on average does the processor execute *between* each pair of conditional branches?

1.2 Part 2: Comparing ALPHA and PISA ISAs

In this part, you will execute the same set of smaller test programs on both Alpha and PISA configurations to compare the resulting instruction counts and distributions.

1.2.1 Section A: Running the ALPHA Benchmarks

Ensure your simulator is still configured for **ALPHA**. If not, repeat Step 1 from Part 1. Run the following commands and record the results in Table 2.

```
./sim-profile -iclass ./tests-alpha/bin/test-math
./sim-profile -iclass ./tests-alpha/bin/test-fmath
./sim-profile -iclass ./tests-alpha/bin/test-llong
./sim-profile -iclass ./tests-alpha/bin/test-printf
```

Table 2: ALPHA ISA Comparison Results

Benchmark	Total Insts	Load %	Store %	Uncond. Branch %	Cond. Branch %	Integer Compute %	Floating Pt. Compute %
test-math							
test-fmath							
test-llong							
test-printf							

1.2.2 Section B: Running the PISA Benchmarks

Step 1: Re-configure for PISA Now, change your simulator configuration to **PISA**.

```
make clean
make config-pisa
make
```

Step 2: Run the PISA Benchmarks Execute the following commands. Note the new path to the PISA binaries. Record the results in Table 3.

```
./sim-profile -iclass ./tests-pisa/bin.little/test-math
./sim-profile -iclass ./tests-pisa/bin.little/test-fmath
./sim-profile -iclass ./tests-pisa/bin.little/test-llong
./sim-profile -iclass ./tests-pisa/bin.little/test-printf
```

Table 3: PISA ISA Comparison Results

Benchmark	Total Insts	Load %	Store %	Uncond. Branch %	Cond. Branch %	Integer Compute %	Floating Pt. Compute %
test-math							
test-fmath							
test-llong							
test-printf							

1.2.3 Final Analysis

Compare the results from the Alpha and PISA runs. Create a histogram (using MATLAB, Excel, or another tool) comparing a key metric, such as the total number of instructions for each benchmark across the two ISAs.

- What can you conclude about the two ISAs from your data and plot? For instance, is one ISA more "dense" than the other (i.e., does it accomplish the same task with fewer instructions)?